

INFORMATION SYSTEM DEPARTMENT
INFORMATION TECHNOLOGY FACULTY– HCM UNIVERSITY SCIENCE

CSC10006

Introduction to Database

Chapter 00

Introduction to Course

Instructor - Dr. NGUYEN TRAN MINH THU



fit@hcmus



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

Course Information

- ☐ Course ID: **CSC10006**
- ☐ Course Name: **Introduction to Database**
- ☐ Number of Credit: **4**
- ☐ Major: **Information System**
- ☐ Course webpage (for course materials)
 - ☐ Link: <http://docs.ntmthu.net:8080> or <https://docs.ntmthu.net:4343>
 - ☐ User: **db**
 - ☐ Password: **Check the password information on Moodle**
 - ☐ Password (to open document/slide): **ntmthu@db**
- ☐ Class Schedule:
 - ☐ **Friday, 13:30 – 17:10 PM (Room I.44)**

Instructor information

□ Instructor:

- Dr. NGUYỄN TRẦN MINH THƯ' (ntmthu@fit.hcmus.edu.vn)

□ Teaching Assistant (TA)

- MSc. NGUYỄN NGỌC MINH CHÂU (nnmchau@fit.hcmus.edu.vn)

□ Lab Tutor:

- MSc. HỒ THỊ HOÀNG VY (hthvy@fit.hcmus.edu.vn)
- MSc. NGUYỄN NGỌC MINH CHÂU (nnmchau@fit.hcmus.edu.vn)

□ Communication channels: Moodle, Facebook Group

Classroom Etiquette



RESPECTFUL

- **BE ON TIME**
- **MOBILE PHONE ON SILENT**
- **DON'T DISTRACT YOUR CLASSMATES**

COLLABORATIVE

- **JOIN TEAMWORK**
- **SHARE YOUR IDEAS**
- **RAISE QUESTIONS**

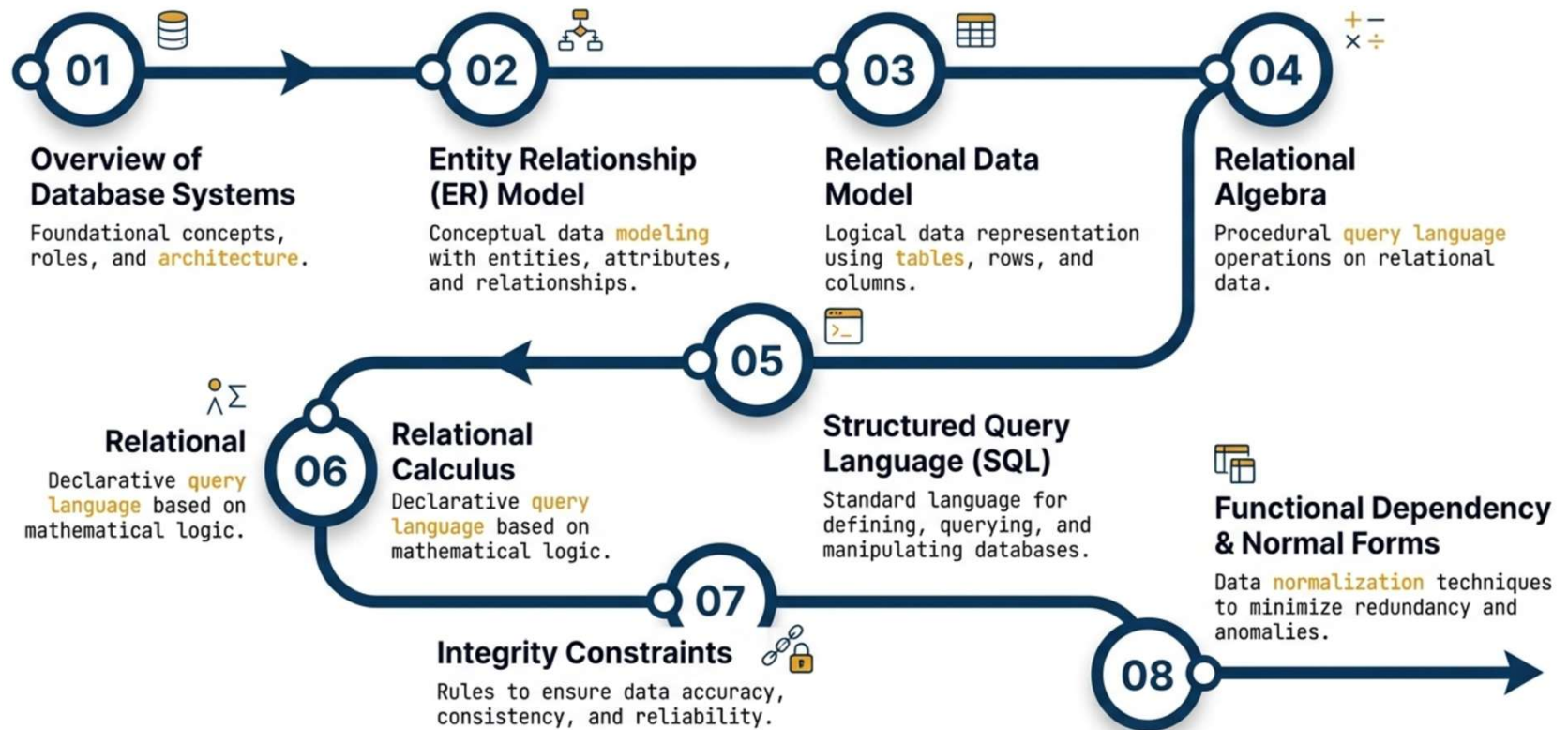
Course Objectives

- Explain why databases matter in organizations and describe the core concepts of databases and DBMS.
- Use data models, especially entity relationship model (ERM) and relational data model (RDM).
- Identify and define key data constraints, including keys and integrity constraints.
- Practice the database query languages, including Relational Algebra, SQL, Relational Calculus.
- Understand the functional dependency, normal forms and normalization
- Evaluate relational database schemas at a basic level, including consistency and normalization.
- Create ER diagrams and relational database schemas for simple, real-world scenarios.
- Use the DBMS (Microsoft SQL Server) to build a relational database schema and apply SQL to manipulate and query data (INSERT, UPDATE, DELETE, and SELECT).

Course Outline

- ☐ Chapter 01: An overview of database systems
- ☐ Chapter 02: Entity relationship model
- ☐ Chapter 03: Relational data model
- ☐ Chapter 04: Relational Algebra
- ☐ Chapter 05: Structured Query Language (SQL)
- ☐ Chapter 06: Relational Calculus
- ☐ Chapter 07: Integrity constraints
- ☐ Chapter 08: Functional dependency and Normal forms

Course Roadmap



Teaching Plan

Week	Topic	Lecturer	Slide
1	An overview of database systems Relational data model (RDM)	NTMThu	Chapter 01 & 03
2	Relational algebra (RA) (Part I) & SQL (Part I)	NTMThu	Chapter 04 & 05
3	Relational algebra (Part II) & SQL (Part II)	NTMThu	Chapter 04 & 05
4	Relational algebra (Part III) & SQL (Part III)	NTMThu	Chapter 04 & 05
5	TA: Activities & Practice (RDM, RA & SQL)	NNMChau	Chapter 04 & 05
6	TA: Activities & Practice (RA & SQL)	NNMChau	Chapter 04 & 05
7	TA: Activities & Practice (RA & SQL)	NNMChau	Chapter 04 & 05
8	Relational Calculus & Integrity constraints	NTMThu	Chapter 06 & 07
9	FD, NFs, and Normalization (Part I)	NTMThu	Chapter 08
10	FD, NFs, and Normalization (Part II) Entity Relationship Model	NTMThu	Chapter 08 & 02
11	Review Activities & Practice	NTMThu	

Course Assessment

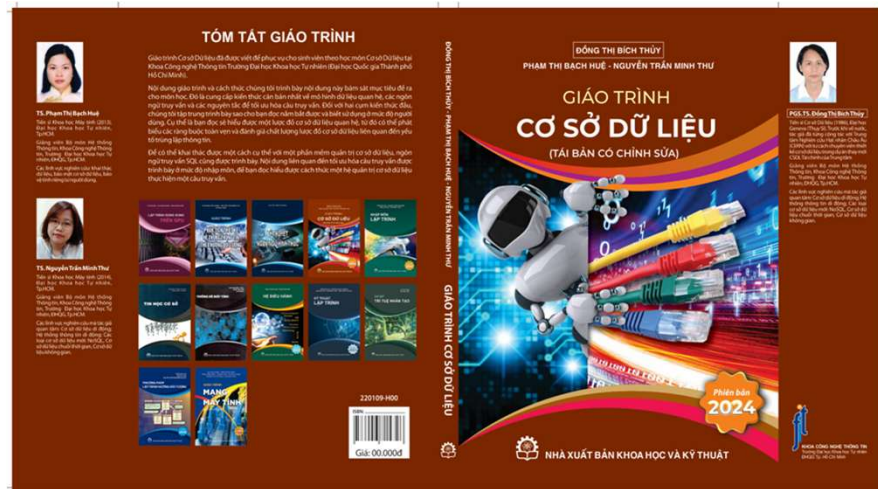
ID	Categories	Description	Ratio (%)
A1	Theory Process Score	Small quizzes in class for each topic Exercises in class for each topic Practicing based on knowledge taught in class (homework) Test (midterm)	20%
A2	Lab Process Score	Lab exercises	30%
A3	Final Exam	Written exam, Closed book exam. Describe the understanding of different topics, analyze & program to solve problems	50%
TOTAL			100%

Bonus Opportunity: 10%

Reference Materials

Primary Textbook:

Giáo trình Cơ sở dữ liệu, Đồng Thị Bích Thủy, Phạm Thị Bạch Huệ, Nguyễn Trần Minh Thư Nhà xuất bản Khoa học kỹ thuật, 2010 (tái bản 2015).



Supplementary Reference

- ☐ Coronel/Morris/Rob: **Database Systems: Design, Implementation, and Management**, 13e, ISBN-13: 978-1-111-96960-8
- ☐ Ramez Elmasri, ShamkantB. Navathe, **Fundamentals of Database Systems**, 7th Edition, Pearson Education, 2016.
- ☐ Abraham Silberschatz, Henry F. Korth, S. Sudarshan, **Database System Concepts**, 6th Edition, McGraw-Hill, 2006
- ☐ Jeffrey A. Hoffer, Mary B. Prescott and Heikki Topi. **Modern Database Management**, 10th Edition, Pearson Education Limited 2016
- ☐ Ullman and Widon, **A first course in database systems**, 3rd rd Edition, Prentice Hall, 1997

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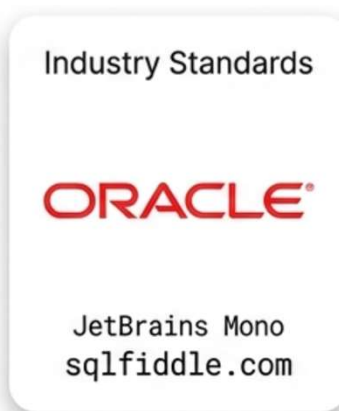
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Link: <https://mooc.vnuhcm.edu.vn/>

Technical Toolkit



JetBrains Mono
sqlfiddle.com

